

Case Study: Cycling Traffic Data in Flanders

Group 4

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Contents

1	Introduction	1
1.1	Background and Motivation	1
1.2	Problem Statement	1
1.3	Research Objectives	1
2	Data Sources and Preprocessing	2
2.1	Data Sources	2
2.1.1	Cycling Traffic Data (Fietstellingen)	2
2.1.2	Historical Accident Data	2
2.1.3	Weather Data (Open-Meteo)	2
2.2	Preprocessing Pipeline	2
2.2.1	Temporal Alignment	2
2.2.2	Geographic Filtering and Category Refinement	3
2.2.3	Spatial Buffer Join	3
2.2.4	Station Grouping	3
3	Exploratory Data Analysis	3
3.1	Traffic Patterns	3
3.2	Accident Characteristics	4
3.3	Environmental Conditions at Time of Accident	5
4	Methodology	5
4.1	Data Aggregation and Monthly Dataset Construction	5
4.2	Target Variable Definition	6
4.3	Feature Engineering	6
4.4	Model Selection and Training	7
4.5	Threshold Tuning	7
4.6	Evaluation Metrics	7
5	Results	8
5.1	Model Comparison	8
5.2	Final Model Performance	8
5.3	Feature Importance Analysis	9
5.4	Historical Safety Rankings	10
5.5	Forecast Safety Rankings	10
6	Web Application	11
6.1	Interactive Risk Map	11
6.2	Historical and Forecast Safety Rankings	12

6.3	Station-Group-Level Detail Panel	12
6.4	Prediction Engine	12
7	Discussion and Conclusion	13
7.1	Interpretation of Results	13
7.2	Limitations	13
7.3	Conclusion and Future Work	14
	Appendices	15
A	GitHub Repository and Application	15
B	Environmental Setup and Key Dependencies	15
C	Full Feature List	15
D	Additional Figures	18

1 Introduction

1.1 Background and Motivation

Cycling is a cornerstone of sustainable mobility in Flanders, yet it remains disproportionately dangerous. In Flanders, cyclists accounted for 30% of all road fatalities in 2024, the second largest group after car occupants, while fatalities among all other road user types have declined significantly since 2005, cyclist deaths have increased by 15% over the same period [Statistiek Vlaanderen, 2024].

To monitor cycling infrastructure, the Flemish road agency Agentschap Wegen en Verkeer (AWV) operates 151 automatic bicycle counting stations across Flanders (*fietstellingen*), recording cyclist counts at 15-minute intervals. AWV also maintains a *Gevaarlijke Punten* (dangerous points) list identifying locations with a history of serious accidents [Agentschap Wegen en Verkeer, 2025]. However, this list is inherently reactive: a location must first accumulate recorded accidents before being flagged, and it does not account for cycling exposure, a busy junction with ten accidents may be safer per cyclist than a quiet path with two.

1.2 Problem Statement

This project addresses that gap by developing a predictive cycling risk assessment model for AWV sensor locations. The model uses lagged traffic volume, weather conditions, and accident history from the preceding months to forecast whether a location will experience elevated risk in the following month. Because all features are drawn from the period prior to the prediction window, the model is genuinely forward-looking and avoids data leakage from concurrent observations. The analysis is scoped to 87 grouped station entities derived from 151 individual AWV EcoCounter sensors, which provide the exposure data necessary to compute normalized risk estimates.

1.3 Research Objectives

1. Produce reliable next-month crash risk predictions for AWV station groups across Flanders.
2. Generate historical and forward-looking safety rankings, identifying station groups where accident risk is disproportionate to cycling exposure.
3. Develop an interactive web dashboard that visualises historical and forecast safety rankings across AWV station groups, and provides a station-level prediction engine allowing users to adjust traffic and weather inputs to simulate next-month crash risk.

2 Data Sources and Preprocessing

2.1 Data Sources

2.1.1 Cycling Traffic Data (Fietstellingen)

The *Fietstellingen* dataset comprises monthly count records (`data-YYYY-MM.csv`) containing 15-minute interval cyclist counts per sensor, a site metadata file (`sites.csv`) providing the geographic coordinates, municipality, and installation date of each EcoCounter station, and a directional configuration file (`richtingen.csv`).

2.1.2 Historical Accident Data

The *Geolocation of Road Traffic Accidents 2017–2024* dataset published by Statbel [Statbel, 2025] records all police-reported road traffic accidents in Belgium with geographic coordinates. Each record includes the year and month of collision, road user types, weather and road conditions at the time of the accident, and the municipality of occurrence.

2.1.3 Weather Data (Open-Meteo)

Hourly weather observations for each sensor location were retrieved from the Open-Meteo Historical Archive API [Open-Meteo, 2024] using each sensor’s geographic coordinates, covering the period from its installation date onwards. Variables include temperature, relative humidity, precipitation, snowfall, cloud cover, wind speed, wind direction, and weather code. Data are retrieved at the nearest available ERA5 reanalysis grid point, which has a spatial resolution of approximately 5km; sensor locations within this distance of each other therefore share identical weather observations.

2.2 Preprocessing Pipeline

Data quality checks identified three sensors requiring special treatment: sensor 144 recorded only one month of data with 100% missing counts, sensor 142 was identified as a test sensor with a 10.9% null rate, and both were excluded from analysis. Sensor 123 exhibited a null rate of 76.8% and is retained but treated with caution. A spike in missing values between March and July 2022 suggests a network-wide data collection issue; affected records are excluded via null dropping.

2.2.1 Temporal Alignment

All data sources are aligned to the period August 2019 to December 2024. The lower bound corresponds to the earliest available fietstellingen count data in this study, while the upper bound is determined by the coverage of the Statbel accident dataset, which extends to December 2024.

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2.2.2 Geographic Filtering and Category Refinement

Accident records are filtered to retain only those involving at least one cyclist as a road user and falling within the geographic bounding box of the AWW sensor network.

2.2.3 Spatial Buffer Join

Accident coordinates, originally provided in the Belgian Lambert projection (EPSG:31370), are converted to WGS84 (EPSG:4326) prior to spatial joining. Of the filtered accident records, 10.3% lack valid coordinate information and are excluded from the spatial join.

The remaining records are matched to sensors using a BallTree radius search within a 1km threshold. A threshold of 1km was selected over the stricter 500m radius, as the smaller radius yielded insufficient crash-sensor matches for reliable monthly modelling. This choice is further supported by model performance: the 1km dataset yields a ROC AUC of 0.749 compared to 0.715 at 500m, and a substantially higher F1 score (0.579 vs 0.381), attributed to the more balanced class distribution at 1km (30.1% positive) compared to 500m (13.4% positive).

2.2.4 Station Grouping

As the *fietstellingen* network frequently deploys multiple sensors at the same physical location, such as separate inbound and outbound counters at the same junction, individual sensors are consolidated into station groups, forming the unit of analysis for all subsequent modelling. Two sensors are merged into the same group if they share the same municipality, have matching normalized names, and are located within 200 metres of each other. Traffic and crash features are aggregated at the group level, with the group’s geographic position set to the centroid of its member sensors. Accident records are deduplicated within each group to ensure that a single crash falling within range of multiple member sensors is counted only once. This process reduces the 151 individual sensors to 95 grouped station entities.

3 Exploratory Data Analysis

3.1 Traffic Patterns

Figure 1 shows the distribution of total cyclist counts by hour of day and day of week across all sensors and the full study period. Cycling activity exhibits two clear peaks: a morning peak at 08:00 and an afternoon peak at 16:00–17:00, consistent with commuting behaviour. Weekday traffic is substantially higher than weekend traffic, with Tuesday and Wednesday recording the highest counts, reflecting the dominance of commuting trips in the dataset.

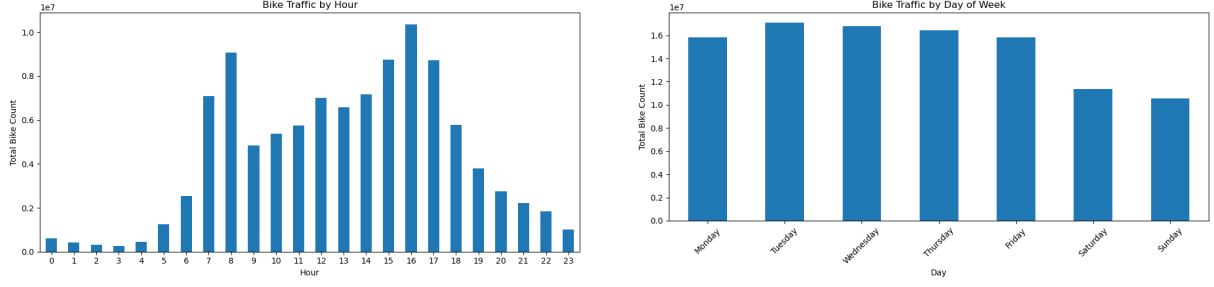


Figure 1: Total cyclist counts by hour of day (left) and day of week (right) across all sensors (2019–2024).

3.2 Accident Characteristics

Figure 2 shows the temporal distribution of bicycle accidents in Flanders across year, month, and hour of day.

Annual accident counts remain broadly stable between 2017 and 2024, ranging from approximately 6,500 to 8,400 per year, with a notable peak in 2022 and 2023. The absence of a clear downward trend suggests that cycling safety in Flanders has not improved significantly over the study period despite broader road safety efforts.

Monthly distribution reveals a strong seasonal pattern, with accidents peaking in June and September and reaching their lowest levels in January and February. This reflects the higher cycling activity during warmer months and supports the inclusion of seasonality features in the model.

The hourly distribution mirrors the commuting pattern observed in traffic flow, with clear peaks at 08:00 and 16:00–17:00. This alignment between high traffic volume and high accident frequency during peak commuting hours underscores the relevance of traffic volume as a predictor of crash occurrence.

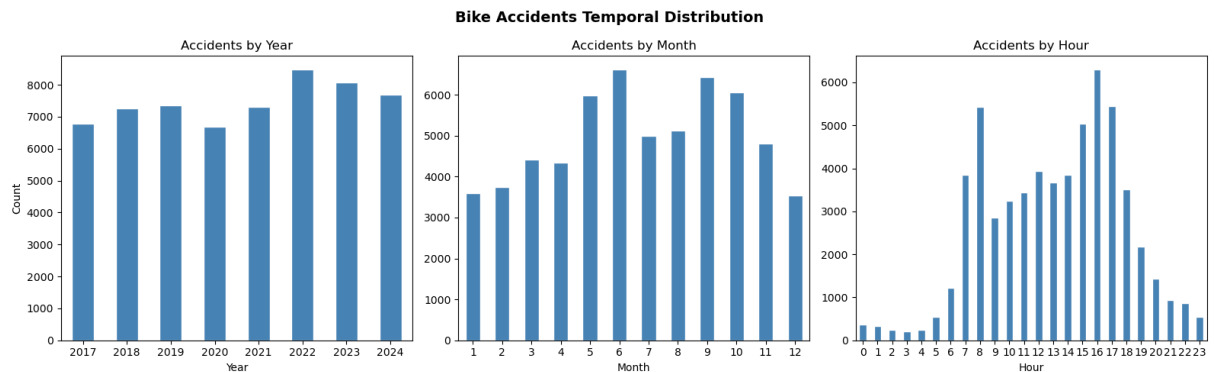


Figure 2: Temporal distribution of bicycle accidents in Flanders by year, month, and hour of day (full dataset 2017–2024; analysis period restricted to 2019–2024).

3.3 Environmental Conditions at Time of Accident

Figure 3 shows the distribution of weather and road surface conditions at the time of accident. The majority of accidents occurred under normal weather (83.1%) and on dry road surfaces (75.5%). Although these distributions may appear counter-intuitive, they likely reflect higher cycling exposure under favorable conditions rather than lower inherent risk during adverse weather or wet roads. As adverse conditions account for a non-trivial share of accidents, 6.8% during rainfall and 13% on wet road surfaces, weather variables are included as features in the predictive model.

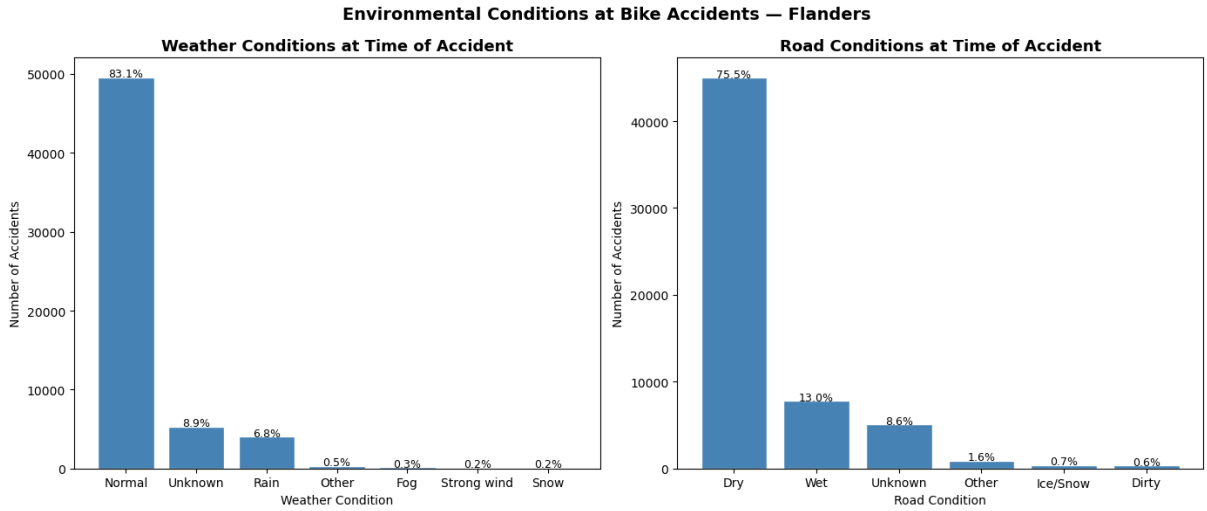


Figure 3: Weather and road surface conditions at the time of bicycle accident in Flanders (2017–2024).

Regarding light conditions, 79% of accidents occurred during daylight, consistent with the hourly traffic and accident patterns identified in Sections 3.1 and 3.2, where both cycling volume and accident frequency peak during daytime commuting hours.

Accidents are roughly equally distributed between intersection (44.5%) and non-intersection locations (55.5%).

4 Methodology

4.1 Data Aggregation and Monthly Dataset Construction

The unit of analysis for this study is the station-month: one row per grouped station entity per calendar month. Three data streams are independently aggregated to this level before being merged into a single modelling dataset.

Traffic data are aggregated from 15-minute interval counts to monthly summaries per station and direction (inbound and outbound), yielding features including total cyclist count, mean and maximum interval counts, and the number of zero-count intervals.

Hourly weather observations are aggregated to monthly summaries per station, producing mean, minimum, and maximum temperature, total precipitation, total snowfall, average cloud cover, average wind speed, and event-based counts including rainy days, heavy rain days, frost days, and storm days.

Matched accident records are counted per station per month to produce two crash outcome variables: `bike_crash_count_1km_grouped`, the number of bicycle accidents matched within 1km of the station in that month, and `bike_crash_happened_1km_grouped`, a binary indicator of whether at least one such accident occurred.

An inner join between traffic and weather ensures only station-months with complete observations from both sources are retained. The dataset is further restricted to months in which at least one bicycle accident was recorded within the study area; in practice, this condition is satisfied for all months in the 2019–2024 study period. Of the 95 grouped station entities, 87 contribute at least one valid station-month row after requiring complete traffic, weather, and temporal lag features; the remaining 8 are excluded due to insufficient data coverage. The final forecasting dataset comprises 2,733 station-month rows.

4.2 Target Variable Definition

The primary prediction target is a binary indicator of whether at least one bicycle accident occurs within 1km of a station group in the following calendar month:

$$y_{i,t} = \mathbf{1} [\text{bike_crash_count_grouped}_{i,t+1} > 0] \quad (1)$$

where i denotes the station group and t the current month.

Each row in the forecasting dataset represents the state of a station group at month t , with features derived from months prior to t , and the target reflecting outcomes in month $t + 1$. This temporal structure ensures all features are available at prediction time.

4.3 Feature Engineering

For each of the three feature sources, traffic, weather, and accident history, the core monthly values are transformed using one- and two-month lags, three- and six-month rolling means, and three-month rolling standard deviations.

- One- and two-month lags
- Three- and six-month rolling means
- Three-month rolling standard deviation

All transformations use only values from months prior to the prediction month to prevent data leakage.

Traffic features include monthly inbound and outbound cyclist counts, mean interval counts, zero-count intervals, month-on-month changes in flow, inbound-to-outbound ratio,

and expanding historical averages per station group.

Weather features include monthly temperature, precipitation, wind speed, and cloud cover, along with month-on-month changes in temperature and precipitation. Additionally, daily weather event counts are included: number of rainy days, heavy rain days, frost days, and storm days per month.

Accident history features include the grouped monthly crash count within 1km, a binary indicator of crash occurrence, crash rate per 1,000 cyclists, three- and six-month rolling crash sums, and an expanding historical average crash count per station group.

Seasonality is encoded using sine and cosine transformations of the month of year to capture its cyclical nature.

4.4 Model Selection and Training

Several classifiers were evaluated on the grouped station dataset, including Random Forest, XGBoost, LightGBM, and CatBoost. CatBoost was selected as the final model based on its superior ROC AUC and F1 score on the test set, as reported in Section 5.1.

A three-way time-based split is applied: the earliest 60% of months form the training set (September 2019 – September 2022), the following 20% form the validation set (October 2022 – September 2023), and the remaining 20% form the held-out test set (October 2023 – December 2024). Hyperparameter tuning is performed on the validation set, and the final model is retrained on the combined training and validation data using the optimal parameters before evaluation on the held-out test set.

4.5 Threshold Tuning

CatBoost outputs a predicted probability of crash occurrence for each station group. A decision threshold is selected by searching over values from 0.05 to 0.95 in steps of 0.01 and choosing the threshold that maximizes F1 score on the validation set. This approach balances precision and recall, which is preferable to the default threshold of 0.5 given the class imbalance in the dataset (30.5% positive target share). The selected threshold is then applied to the held-out test set without further adjustment, ensuring an unbiased final evaluation.

4.6 Evaluation Metrics

Model performance is evaluated using the following metrics on the held-out test set:

- **ROC AUC:** measures the model’s ability to rank high-risk station-months above low-risk ones, independent of the classification threshold.
- **F1 score:** the harmonic mean of precision and recall, capturing the balance between false positives and false negatives at the selected threshold.

- **Precision:** the proportion of predicted high-risk station-months that are truly high-risk.
- **Recall:** the proportion of truly high-risk station-months that are correctly identified by the model.
- **Confusion matrix:** a breakdown of true positives, false positives, true negatives, and false negatives at the selected threshold.

Accuracy is also reported; however, it is not the primary criterion given the minority positive class.

5 Results

5.1 Model Comparison

Figure 4 compares the test-set performance of four classifiers with default hyperparameters. Although performance differences are modest, CatBoost achieves the highest ROC AUC (0.749) and F1 (0.579) and is therefore selected for further tuning. Its gradient boosting architecture also offers native support for categorical features and missing values, making it well-suited for the heterogeneous station-level dataset.

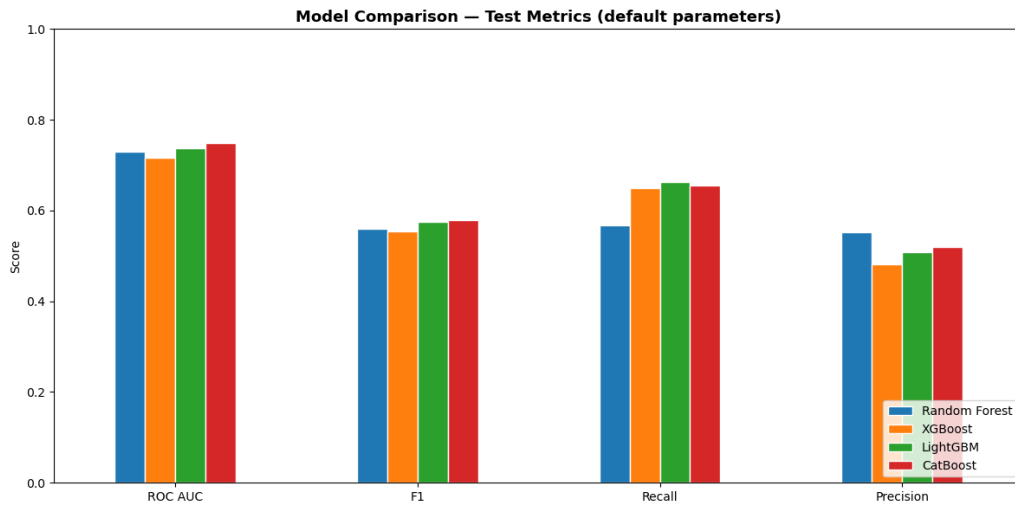


Figure 4: Test-set performance comparison of four classifiers on the grouped station dataset with default hyperparameters.

5.2 Final Model Performance

Following hyperparameter tuning with Optuna and threshold optimisation, the final CatBoost model achieves a test ROC AUC of 0.774, F1 of 0.598, recall of 0.667, and precision of 0.542 at a classification threshold of 0.32 (Table 1). The model correctly identifies 66.7% of high-risk station-months, with 54.2% of predicted high-risk cases being true positives.

Metric	Validation	Test
ROC AUC	0.780	0.774
F1	0.581	0.598
Recall	0.822	0.667
Precision	0.450	0.542
Accuracy	0.650	0.729

Table 1: Final CatBoost model performance after hyperparameter tuning and threshold optimisation (threshold = 0.32).

Performance on the test set is consistent with validation results, with test F1 (0.598) slightly exceeding validation F1 (0.581), suggesting the model generalizes well to unseen data.

Figure 5 shows the confusion matrices for both the validation and test sets, illustrating the distribution of true positives, false positives, true negatives, and false negatives at the selected threshold.

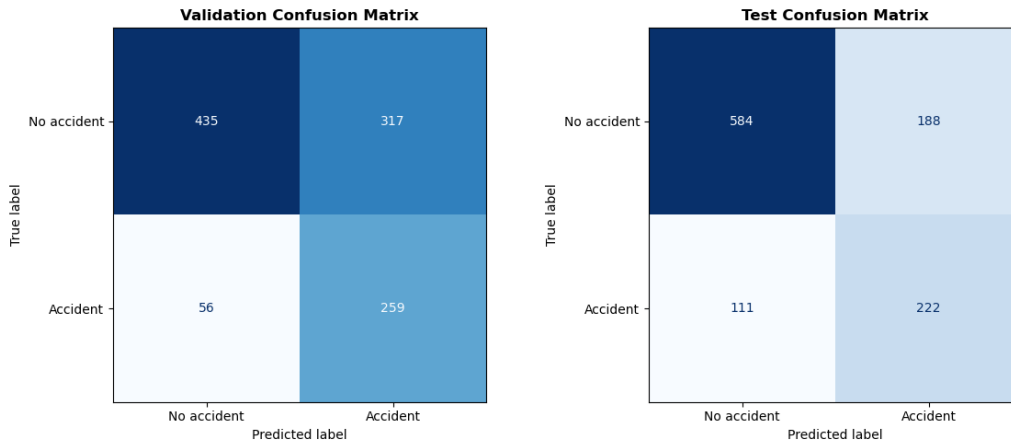


Figure 5: Confusion matrices for the final CatBoost model on the validation set (left) and test set (right).

5.3 Feature Importance Analysis

Figure 6 shows the top 20 features by importance score from the final CatBoost model. Geographic identifiers, municipality (`gemeente`), group name, and latitude, rank among the most important features. The historical average crash count per station group (`group_avg_bike_crash_count_history`) is the second most important feature. Among weather features, variability measures such as rolling standard deviations of cloud cover, precipitation, and rainy days feature prominently.

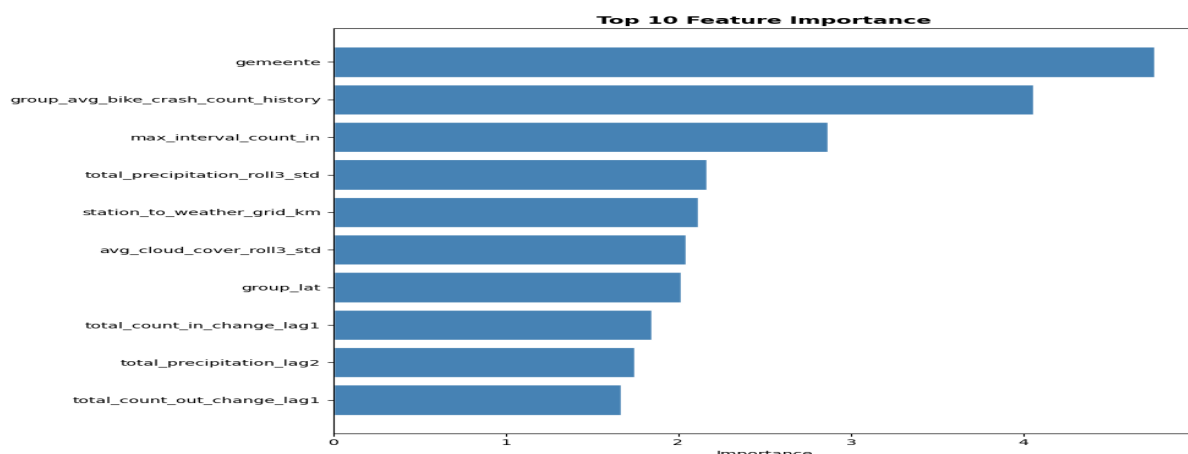


Figure 6: Top 10 feature importance scores from the final CatBoost model.

5.4 Historical Safety Rankings

Table 2 presents the top 10 station groups ranked by smoothed historical crash rate per 10,000 cyclists, providing a retrospective view of observed safety performance over the study period.

Rank	Station Group	Municipality	Total Crashes	Smoothed Rate
1	Genk	Genk	64	1.941
2	St. Pieters Leeuw teller 2	Halle	4	0.885
3	St. Pieters Leeuw teller 1	Sint-Pieters-Leeuw	4	0.744
4	Kortrijk teller 1	Kortrijk	40	0.707
5	Maasmechelen teller 1	Maasmechelen	19	0.620
6	Bree	Bree	50	0.517
7	Asse teller 2	Asse	4	0.511
8	Pelt teller 1	Pelt	22	0.481
9	Gent	Gent	69	0.472
10	Maasmechelen teller 2	Maasmechelen	20	0.470

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Table 2: Top 10 station groups by smoothed historical crash rate per 10,000 cyclists (2019–2024).

Genk records the highest smoothed crash rate (1.941 per 10,000 cyclists), more than twice the rate of the second-ranked group. Maasmechelen appears twice in the top 10, suggesting a concentration of risk within that municipality.

5.5 Forecast Safety Rankings

Of the 87 station groups, 23 are predicted as high risk for the following month at the selected threshold of 0.32. Figure 7 identifies station groups predicted as high risk that do not appear in the official *Gevaarlijke Punten* list, representing locations where the model identifies elevated risk that has not yet been formally recognized.

Kraainem–Tervuren receives the highest predicted probability (0.85), followed by Nieuwpoort teller 1 (0.76) and Tienen 2 (0.74). These locations warrant attention from AWW planners as potential emerging risk spots.

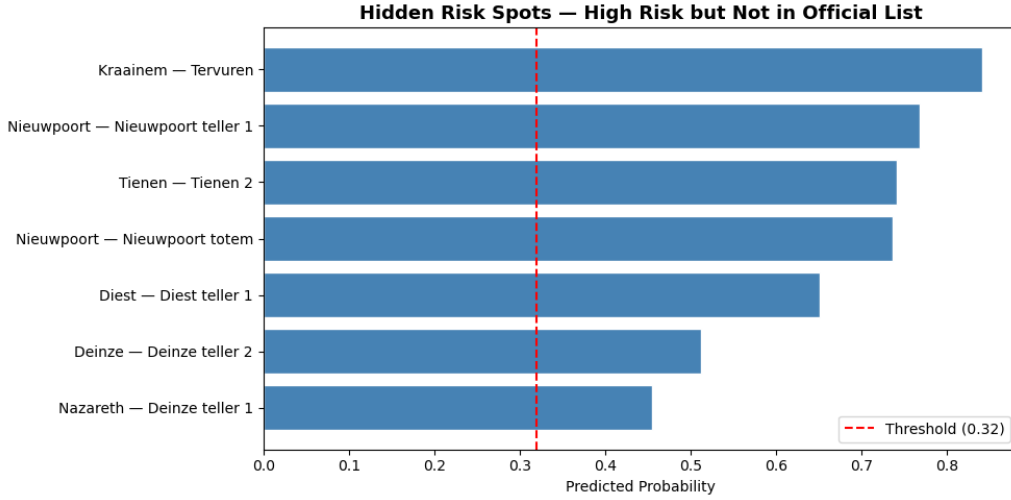


Figure 7: Station groups predicted as high risk (probability ≥ 0.32) that do not appear in the official Gevaarlijke Punten list.

6 Web Application

6.1 Interactive Risk Map

Figure 8 shows the interactive map displaying all 87 station groups across Flanders, colour-coded by predicted crash probability. Each marker displays the group's risk rank under the currently selected ranking mode (historical or forecast), with hidden risk spots additionally marked with an exclamation mark. Clicking on any marker reveals the group's forecast probability, historical rank, and municipality.

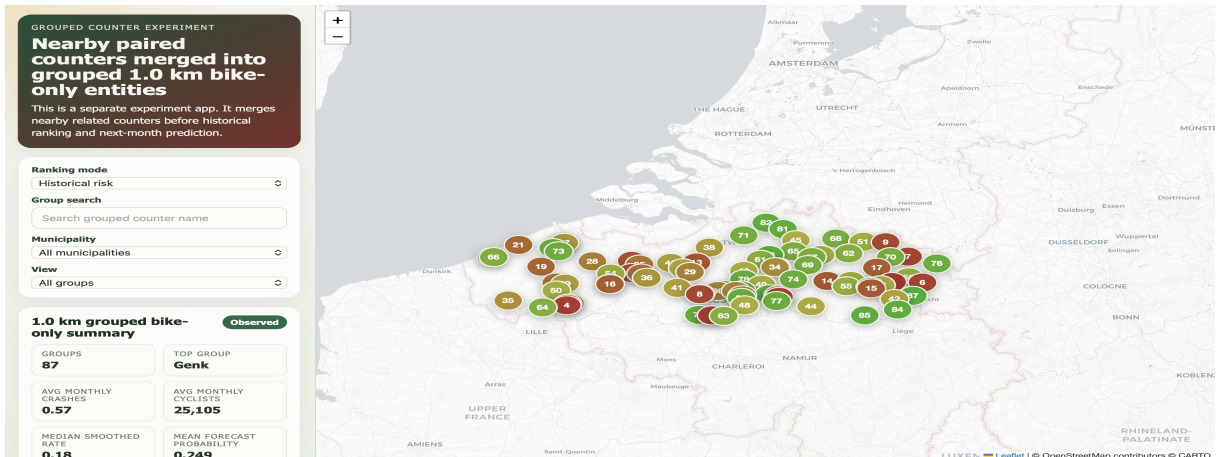


Figure 8: Interactive risk map showing all station groups across Flanders, colour-coded by forecast crash probability.

6.2 Historical and Forecast Safety Rankings

The application provides two ranking modes accessible from the left panel. The historical ranking orders station groups by smoothed crash rate per 10,000 cyclists over the study period, reflecting observed safety performance. The forecast ranking orders groups by predicted crash probability for the following month, enabling forward-looking prioritisation of safety interventions. Users can switch between the two modes to compare historical and predicted risk profiles across station groups.

6.3 Station-Group-Level Detail Panel

Selecting a station group on the map opens a detail panel displaying both historical and forecast information (Figure 9). Historical statistics include total crashes, total cyclists, crash rate per 10,000 cyclists, smoothed risk score, and historical risk rank. Forecast information includes the predicted crash probability, binary risk label, and forecast risk rank. Average monthly statistics and the baseline month used for prediction are also displayed.

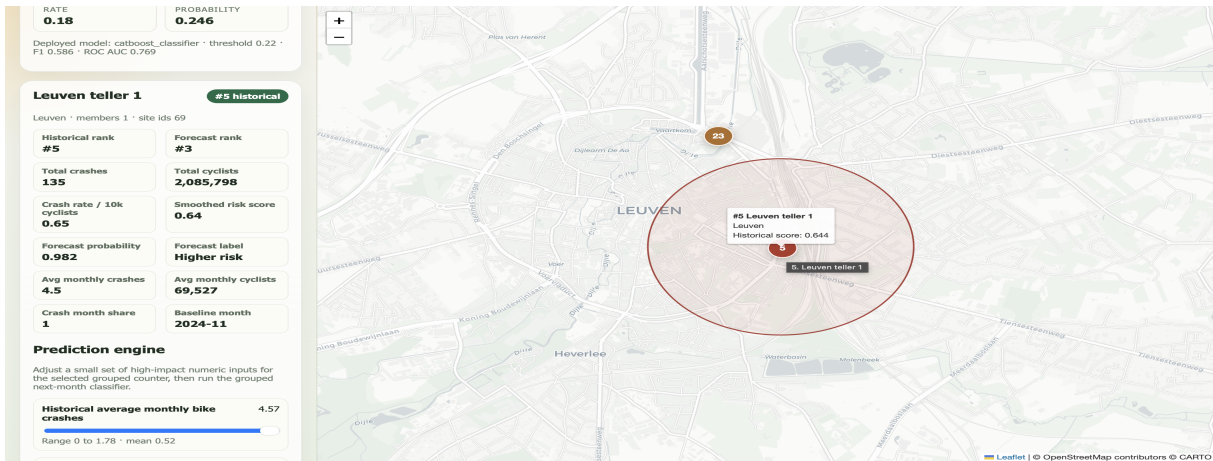


Figure 9: Station-group detail panel showing historical and forecast risk information for the selected group.

6.4 Prediction Engine

The prediction engine allows users to adjust key input features for a selected station group using sliders, then run the deployed CatBoost classifier to obtain a next-month crash probability. Adjustable inputs include historical average monthly crashes, previous month traffic volume (inbound and outbound), crash rate per 1,000 cyclists, precipitation, wind speed, cloud cover, and recent crash counts over three- and six-month windows. The engine returns a predicted probability and binary risk label at the selected threshold of 0.32, enabling planners to simulate how changes in traffic or weather conditions would affect predicted risk.

7 Discussion and Conclusion

7.1 Interpretation of Results

The final CatBoost model achieves a test ROC AUC of 0.774 and F1 of 0.598, indicating a meaningful ability to distinguish high-risk from low-risk station-months beyond random chance. The relatively high recall (0.667) at the selected threshold means the model correctly identifies two-thirds of genuinely high-risk months, which is particularly relevant for a safety application where missing a high-risk location carries greater practical cost than a false alarm.

Feature importance analysis reveals that geographic identifiers: municipality and group location are the strongest predictors of crash risk. While traffic volume features contribute to the model, their predictive importance is secondary to geographic and historical crash features, suggesting that location-specific structural factors are more deterministic of cycling risk than exposure alone. The prominence of historical crash count as the second most important feature confirms that past risk is a strong indicator of future risk, consistent with the persistent nature of dangerous road locations.

Among the 23 station groups predicted as high risk, several do not appear in the official *Gevaarlijke Punten* list, representing locations where the model identifies elevated risk that has not yet been formally recognized. These hidden risk spots, including Kraainem–Tervuren (predicted probability 0.85) and Nieuwpoort teller 1 (0.76), may represent emerging or underreported danger zones that warrant attention from AWW road safety planners.

7.2 Limitations

Two primary limitations constrain the scope of this study.

First, the analysis is restricted to the 87 grouped station entities derived from the AWW EcoCounter network, which provides sparse and geographically uneven coverage of Flanders. The majority of recorded bicycle accidents (69.6%) occur within built-up areas, whereas many sensor locations are situated on regional roads outside urban centres. As a result, a large proportion of accidents cannot be spatially matched to any sensor, limiting the model’s coverage to monitored locations only and preventing region-wide risk assessment.

Second, the forecasting pipeline depends on the Statbel accident dataset, which extends only to December 2024. The deployed application therefore generates predictions based on November 2024 as the most recent baseline month, meaning the forecast target is December 2024: a period already elapsed. To produce genuinely forward-looking predictions, the pipeline would need to be updated with more recent sensor and weather data as they become available.

7.3 Conclusion and Future Work

This study presents a predictive cycling safety model for AWV sensor locations in Flanders, combining monthly traffic counts, weather observations, and historical accident records to forecast next-month crash risk at grouped station entities. The deployed CatBoost classifier achieves a test ROC AUC of 0.774 and F1 of 0.598, and identifies 23 station groups as high risk, of which several represent hidden risk spots not currently listed in the official *Gevaarlijke Punten*. These findings demonstrate that a data-driven, exposure-adjusted approach can surface locations that reactive safety assessments may overlook.

Several directions for future work are identified. First, integrating data from municipal cycling counters alongside the AWV network would increase coverage of urban areas, where the majority of accidents occur, improving both model training and deployment coverage. Second, as more recent accident data become available from Statbel, rerunning the pipeline would extend the forecast horizon and enable more timely predictions. Third, incorporating road infrastructure features such as intersection type, cycling lane presence, and speed limits could improve predictive accuracy beyond what traffic and weather alone provide. Finally, expanding the model to a quarterly forecasting horizon may yield more stable predictions, as monthly crash counts at the station level are inherently sparse.

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Appendices

A GitHub Repository and Application

The complete source code is available at: <https://github.com/Ghazal-Sarhadi/modern-data-analytics/tree/final-project>

To run the web application locally:

1. Clone the repository and navigate to the project directory
2. Install dependencies: `uv sync`
3. Start the application: `uv run python src/09_app_grouped.py`
4. Open <http://127.0.0.1:5000> in a browser

B Environmental Setup and Key Dependencies

The project was developed and tested in Python ≥ 3.11 . Dependencies can be installed via `uv sync` using the `pyproject.toml` file in the repository. Key packages and versions used in development are listed below:

- `catboost` 1.2.10, `xgboost` 3.2.0, `lightgbm` 4.6.0 — gradient boosting classifiers
- `optuna` 4.8.0 — hyperparameter tuning
- `flask` 3.1.3 — web application
- `scikit-learn` 1.8.0, `numpy` 2.4.6, `pandas` 3.0.3 — data processing and evaluation
- `geopandas` 1.1.3, `pyproj` 3.7.2 — spatial data processing
- `openmeteo-requests` 1.7.5 — weather data retrieval
- `matplotlib` 3.10.9, `seaborn` 0.13.2 — visualisation

C Full Feature List

All features marked with * are additionally transformed with one- and two-month lags (`_lag1`, `_lag2`), three- and six-month rolling means (`_roll3_mean`, `_roll6_mean`), and three-month rolling standard deviation (`_roll3_std`).

Feature	Source	Description
<i>Traffic features</i>		
<code>total_count_in*</code>	Fietstellingen	Monthly total inbound cyclists

Feature	Source	Description
total_count_out*	Fietstellingen	Monthly total outbound cyclists
mean_interval_count_in*	Fietstellingen	Mean 15-min inbound count
mean_interval_count_out*	Fietstellingen	Mean 15-min outbound count
zero_count_intervals_in*	Fietstellingen	Zero-count intervals (inbound)
zero_count_intervals_out*	Fietstellingen	Zero-count intervals (outbound)
total_count_in_change_lag1	Derived	Month-on-month inbound change
total_count_out_change_lag1	Derived	Month-on-month outbound change
in_out_ratio	Derived	Inbound-to-outbound ratio
in_out_ratio_lag1	Derived	Lagged inbound-to-outbound ratio
total_count_sum_lag1	Derived	Lagged total cyclists
group_avg_total_count_in_history	Derived	Expanding mean inbound cyclists
group_avg_total_count_out_history	Derived	Expanding mean outbound cyclists
<i>Weather features</i>		
avg_temperature_2m*	Open-Meteo	Monthly mean temperature (°C)
total_precipitation*	Open-Meteo	Monthly total precipitation (mm)
avg_wind_speed_10m*	Open-Meteo	Monthly mean wind speed (km/h)
avg_cloud_cover*	Open-Meteo	Monthly mean cloud cover (%)
rainy_days*	Open-Meteo	Number of rainy days per month
heavy_rain_days*	Open-Meteo	Number of heavy rain days
frost_days*	Open-Meteo	Number of frost days
storm_days*	Open-Meteo	Number of storm days
avg_temperature_2m_change_lag1	Derived	Month-on-month temperature change

Feature	Source	Description
total_precipitation_change_lag1	Derived	Month-on-month precipitation change
<i>Accident history features</i>		
bike_crash_count_1km_grouped*	Statbel	Monthly grouped crash count
bike_crash_happened_1km_grouped_lag1	Statbel	Lagged binary crash indicator
bike_crash_happened_1km_grouped_lag2	Statbel	Two-month lagged binary indicator
bike_crashes_per_1000_cyclists_lag1	Derived	Crash rate per 1,000 cyclists
group_avg_bike_crash_count_history	Derived	Expanding mean crash count
<i>Seasonality and spatial features</i>		
month_sin, month_cos	Derived	Cyclical month encoding
group_lat, group_long	Sites	Group centroid coordinates
group_id, group_name	Sites	Group identifier and name
gemeente	Sites	Municipality name
station_to_weather_grid_km	Open-Meteo	Distance to weather grid point

Table 3: Full list of base features used in model training. Features marked with * are additionally transformed with lag and rolling window operations.

D Additional Figures

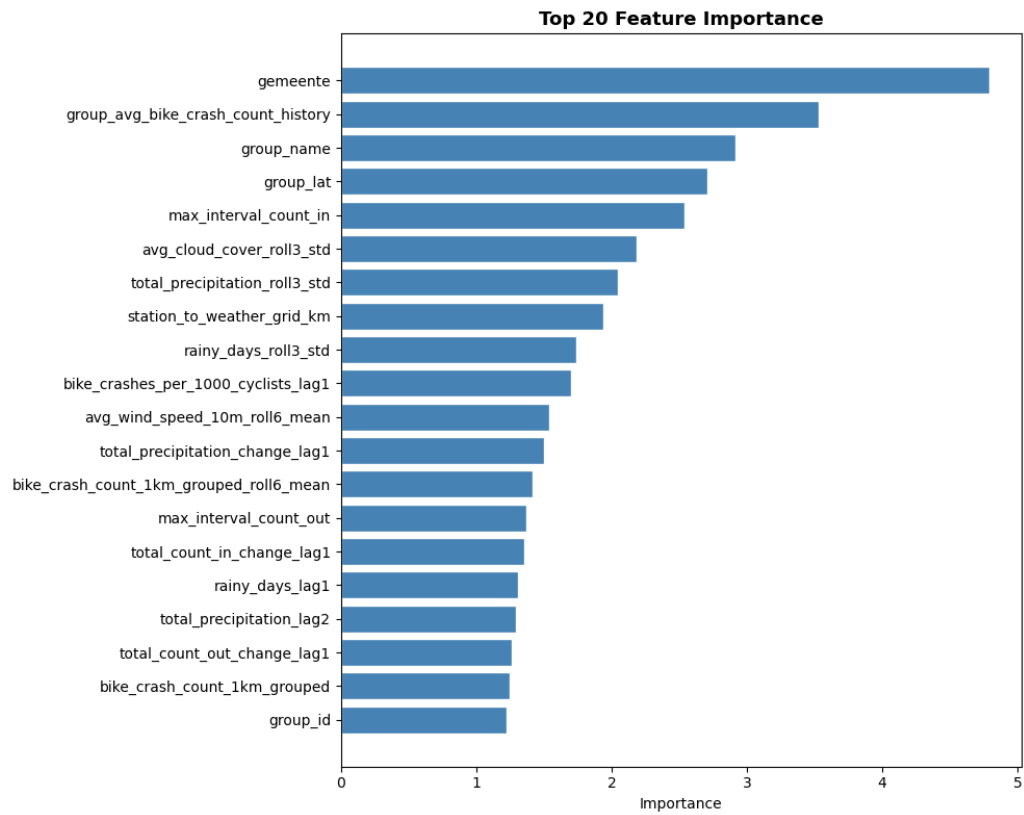


Figure 10: Top 20 feature importance scores from the final CatBoost model.

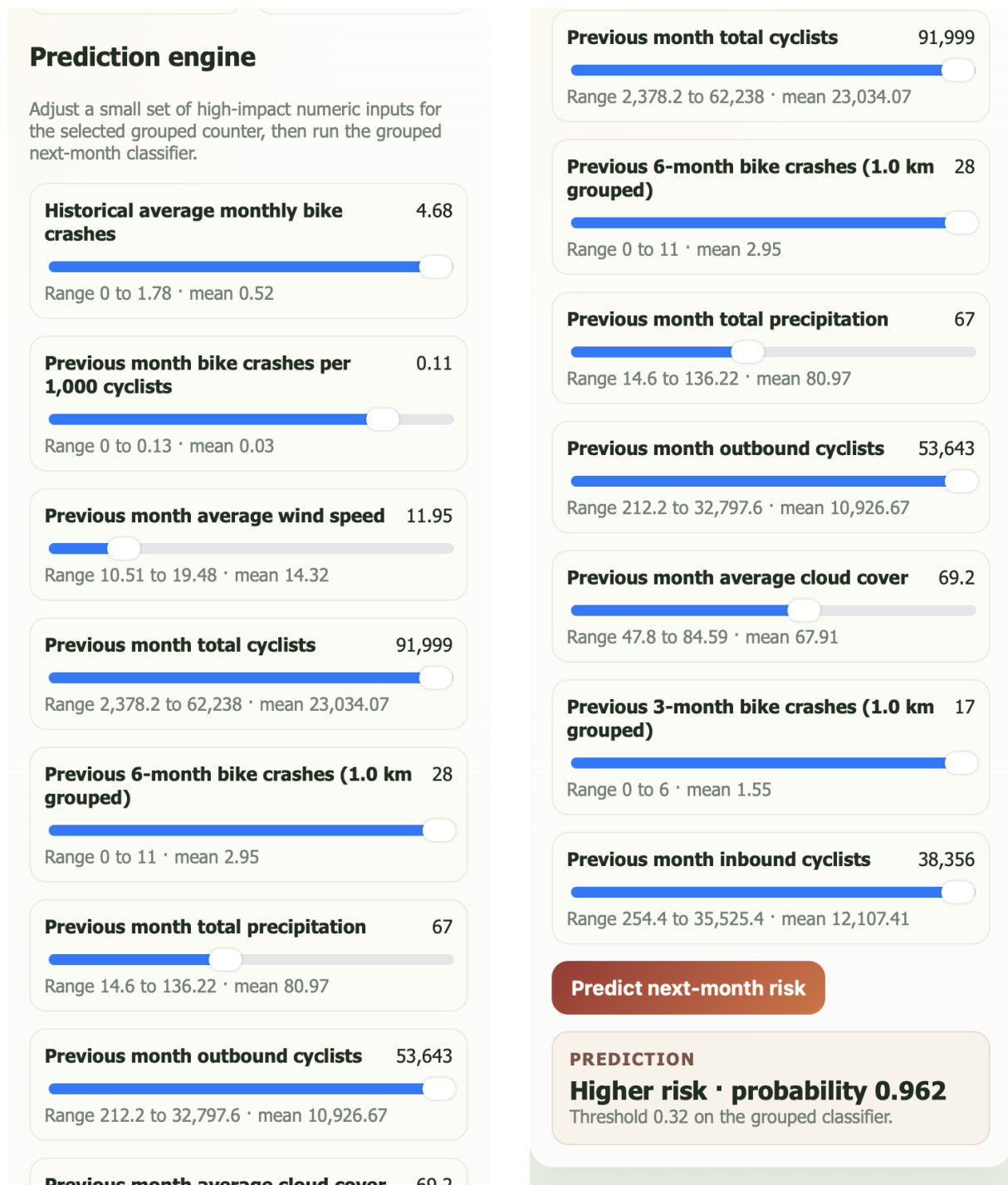


Figure 11: Prediction engine showing adjustable input sliders (left) and the resulting next-month risk prediction (right).